PATRICK BAILLIE ISLES

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SYSTEMS & INFRASTRUCTURE ENGINEER | LINUX KERNEL DIAGNOSTICS | ZERO-ALLOCATION RUST

Systems-focused engineer combining rigorous research methodologies in formal logic, epistemology, and primary-source analysis with low-level systems programming. Proven record diagnosing hardware-to-kernel failure modes, authoring upstream kernel driver tooling, engineering sub-nanosecond telematics engines in Rust, and deploying zero-trust Kerberos/SSSD identity realms on RHEL. Specializes in zero-copy execution, raw IOCTL hardware interrogation, and systematic root-cause debugging.

CORE TECHNICAL COMPETENCIES

Languages:	**Rust** (2021 Edition, Multi-threading, Zero-Copy, FFI), **Python 3** (Typing, AST), **Bash/POSIX Shell**, C (Foundational), SQL
Linux Systems & Kernel:	RHEL/CentOS, Ubuntu, VFS Page Cache Writeback, Raw NVMe IOCTL, SysV IPC, ACPI/C-States, PCIe ASPM, `systemd`, `sysfs` / `procfs`, LVM
Networking & Protocols:	Linux SocketCAN (`AF_CAN`), SAE J1939-21 / CAN 2.0B, UART/Serial (115200 8N1), TCP/IP, Kerberos V5, OpenLDAP (RFC 2307)
Security & Enterprise IAM:	Kerberos KDC (AES-256), SSSD (LDB Dynamic Caching, HBAC), PAM/NSS, SELinux Policies, Kickstart Automation
Observability & Tooling:	Prometheus Metrics Exporters (:9106–:9110), Grafana, Digital Signal Processing (True RMS, DSP), Formal Systems Modeling

SYSTEMS ENGINEERING & OPEN-SOURCE PORTFOLIO

Industrial CAN Telematics & Upstream Linux SocketCAN (can-bus-rs • socketcan-rs PR #107) ***Rust, Linux SocketCAN, SAE J1939***

- Authored official upstream **Pull Request #107** to `socketcan-rs/socketcan-rs` implementing SAE J1939-21 29-bit extended ID decomposition and Parameter Group Number (PGN) parsing; verified clean across all 98 test suites.
- Engineered industrial telematics engine achieving **0.95 ns/frame decode latency** and **1.05+ billion frames/sec throughput** via zero-allocation EFF bitfield extraction over Linux `AF_CAN`.
- Decoded mission-critical PGNs (crankshaft RPM, torque, wheel speed, pneumatic brake pressure) and DM1 Diagnostic Trouble Codes; exposed live metrics via Prometheus (`:9107`) and interactive terminal TUI.

Linux Kernel VFS Page Cache Writeback Choke Auditor (dirty-throttle-sentinel) ***Rust, Linux VFS, Storage IO, Prometheus***

- Engineered a real-time kernel memory auditor interrogating `/proc/sys/vm/` sysctls and `/proc/vmstat` writeback queues to calculate distance-to-throttle headroom before synchronous `TASK_UNINTERRUPTIBLE` (`D` state) write freezes.
- Calculates real-time dirty page accumulation velocity vs. flusher throughput; exposes 15 granular metrics via Prometheus on `:9110` and terminal dashboard with zero heap allocations in the polling loop.

Silicon NVMe 2.0 IOCTL & System V IPC Zombie Memory Sentinels (nvme-sentinel-rs • kmem-drain-rs) ***Rust, Raw IOCTL, SysV IPC, PCIe Telemetry***

- **NVMe Hardware Telemetry**: Interrogated NVMe controllers via raw `NVME_IOCTL_ADMIN_CMD`, decoding 512-byte packed SMART logs, endurance wear, composite temperature, and PCIe ASPM link degradation (`:9108`).
- **Kernel Slab & IPC Reaper**: Interrogated `/proc/sysvipc/shm` and `/proc/slabinfo` to detect dead-PID orphaned shared memory and unreclaimable slab leaks (`SUnreclaim`); implemented safe automated purging via `shmctl(IPC_RMID)` (`:9109`).

High-Performance Embedded DSP Telemetry Engine (telemetry-rs) ***Rust, Embedded UART, Prometheus, DSP***

- Developed multi-threaded, zero-heap UART engine processing 19-byte binary frames with sub-microsecond CRC-16-CCITT validation at **50.79M pkts/sec** with **19.69 ns latency/frame** (Prometheus `:9106`).
- Engineered real-time DSP mathematical core computing True RMS (V_{RMS}), AC peak ripple, and dynamic impedance.

Enterprise Identity Federation & Hardware Root-Cause Analysis (rhel-idm-engine • g14-linux-stability-guide) ***RHEL, Kerberos V5, OpenLDAP, SSSD, ACPI***

- Architected Kerberos V5 (AES-256) / OpenLDAP RFC 2307 realm with SSSD dynamic LDB offline caching and HBAC.
- Isolated AMD Zen 3+ mobile CPU lockups to C6 sleep voltage droop, PCIe ASPM, and NVMe APST collisions; authored automated kernel patch mitigation (`fix-g14.sh`) ensuring 100% uptime.

EDUCATION & FORMAL CREDENTIALS

Electronic Engineering Technology & Mathematics Track • Portland Community College **Honors: Presidents' List (4.0 Term GPA), Multiple Deans' List**

Curriculum & Mastery: Calculus Series (MTH 251: Calculus I, MTH 252: Calculus II), Honors Public Speaking (COMM 111H: A), Technical Writing (WR 227), Electronic Circuit Analysis. Advanced through higher-level calculus with documented academic honors.

Bachelor of Arts (B.A.) in Philosophy • Minor in History (Awarded: March 2020) • Portland State University **International Study: Heidelberg University (Germany)**

International Study: Ruprecht-Karls-Universität Heidelberg • German Language & Continental Philosophy

Advanced Analytical & Research Specialization:

- **Formal Logic & Deduction (PHL 324U):** Propositional/predicate calculus, formal validity, and truth functional systems.
- **Historical Thesis & Research Methodology (HST 492 / HST 495):** 400-level research seminars focused on primary-source critical analysis, historiography, and empirical argumentation.
- **Epistemology & Philosophy of Language (PHL 424 / PHL 433):** Formal models of truth verification, linguistic syntax, semantics, and structural demarcation.
- **German Technical Translation Series (GER 101–301):** Multilingual proficiency in technical German and international academic correspondence.